Figure 1: Operational energy efficiency performance rating scale

3.4 Given the incremental operational carbon intensity reduction factors over time, the boundaries for defining performance ratings should be synchronized accordingly, although the relative distance between the boundaries should not change. The rating of a ship would be determined by the attained CII and the predetermined rating boundaries, rather than the attained CII of other ships. Note that the distribution of ship individual ratings in a specific year

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SOURCE: MEPC.339(76).pdf
determined by the attained CII and the predetermined rating boundaries, rather than the attained CII of other ships. Note that the distribution of ship individual ratings in a specific year may not be always identical with the scenario in 2019, where for example 20% may achieve A, 30% may achieve B, 40% may achieve C, 8% may achieve D and 2% may achieve E in a given year.

4 Method to determine the rating boundaries

4.1 The boundaries can be determined by the required annual operational CII in conjunction with the vectors, indicating the direction and distance they deviate from the required value (denoted as dd vectors for easy reference), as illustrated in figure 2.

Figure 2: dd vectors and rating bands

4.2 Statistically, the dd vectors depend on the distribution of the attained annual operational CII of ships of the type concerned, which can be estimated through a quantile regression, taking data collected through DCS in year 2019 as the sample.

4.3 The quantile regression model for a specific ship type can be developed as follows:

() ()ln( ) ln( ) , {0.15,0.35,0.50,0.65,0.85}ppattained CII c Capacity p= += (5)

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SOURCE: MEPC.339(76).pdf
4.3 The quantile regression model for a specific ship type can be developed as follows:

() ()ln( ) ln( ) , {0.15,0.35,0.50,0.65,0.85}ppattained CII c Capacity p= += (5)
where Capacity is identical with the one used in the operation carbon intensity indicator as specified in the Guidelines on operational carbon intensity indicators and the calculation

methods (G1); p is the typical quantile, meaning the proportion of observations with a lower value is %p ;
()p is the constant term, and ()p is the error term.

4.4 The quantile regression lines in logarithm form are illustrated in Fig.3.

Figure 3: Quantile regression lines in logarithm form

4.5 Then, the dd vectors can be calculated based on the estimates of the intercept (() p), in accordance with Eq.(2), as follows:

(0.15) (0.50)
1
(0.35) (0.50)
2
(0.65) (0.50)
3
(0.85) (0.50)
4

d
d
d
d

=

=

=
=
(6)

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SOURCE: MEPC.339(76).pdf
in accordance with Eq.(2), as follows:

(0.15) (0.50)
1
(0.35) (0.50)
2
(0.65) (0.50)
3
(0.85) (0.50)
4

d
d
d
d

=

=

=
=
(6)

4.6 Through an exponential transformation of each dd vector, the four boundaries fitted in the original data form can be derived based on the required annual operational carbon intensity indicator (required CII), as follows:

1
2
3
4
superior boundary exp( )
lower boundary exp( )
upper boundary exp( )
boundary exp( ) inferior
d required CII
d required CII
d required CII
d required CII
=
= =
=
(7)

Rating boundaries of ship types

The estimated dd vectors after exponential transformation for determining the rating boundaries of ship types are as follows:

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Table 1: dd vectors for determining the rating boundaries of ship types

Ship type
Capacity
in CII
calculation
dd vectors
(after exponential transformation)

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Table 1: dd vectors for determining the rating boundaries of ship types

Ship type
Capacity
in CII
calculation
dd vectors
(after exponential transformation)
exp(d1) exp(d2) exp(d3) exp(d4)